

7.06 Graphical Linear Programming			
7.06a 7.06b	Formulating LP problems	a) Be able to set up a linear programming formulation in the form “maximise (or minimise) objective subject to inequality constraints, and trivial constraints of the form variable ≥ 0”. <i>Includes:</i> 1. <i>identifying relevant variables, including units when appropriate,</i> 2. <i>formulating constraints in these variables, including when the information is given in ratio form,</i> 3. <i>writing down an objective function and stating whether it is to be maximised or minimised.</i>	b) Be able to use slack variables to convert inequality constraints, each being $\leq$ a non-negative constant, into equations, together with further trivial constraints of the form variable ≥ 0 .
7.06c	Working with constraints	c) Be able to investigate constraints and objectives in numerical cases using algebra and ad hoc methods.	
7.06d 7.06e 7.06f	Graphical solutions	d) Be able to carry out and interpret a graphical solution for problems where the objective is a function of two variables, including cases where integer solutions are required. <i>The region where each inequality is not satisfied will be shaded, leaving the feasible region as the unshaded part of the graph.</i>	e) Be able to discuss the effect of making a change to one or two of the coefficients and how this will change the solution. f) Be able to carry out integer programming, including using the branch-and-bound method.